Two Samples of Service Times

Unit 4 · Topic 4.7 · TODAY'S PROBLEM

Suppose two depots independently take SRSs of 36 weekday service-time records from monthly populations of 900 North and 1,200 South records. Technology gives $df = 69.31$ and $t^* = 1.995$ for a 95% two-sample interval.

Tariq: "I sorted both samples and paired equal ranks. The resulting differences have $\bar{d} = 2.2$ and $s_d = 3.0$, giving $(1.18, 3.22)$. Pairing makes the estimate more precise."

Mei: "The records were sampled separately, so I would use a two-sample interval. Sorting observed responses may change the uncertainty rather than reveal design-based pairs."

Service-time samples (min)

Depot	N	n	$\bar{x}$	s
North	900	36	42.3	8.4
South	1,200	36	40.1	7.6

FIRST TAKE (5 min, on your sheet)

Which interval procedure matches the original design? Cite any link that did or did not exist before times were observed.

DOK 1 **(a)** Define $\mu_N - \mu_S$, calculate its point estimate, and name the design-based interval procedure.

DOK 2 **(b)** Verify random, 10 percent, and shape conditions. Compute SE, margin of error, and the 95% interval.

DOK 3 **(c)** Adjudicate the procedures using the separate SRSs, rank pairing, $s_d = 3.0$, and both intervals. Explain how response-based pairing changes uncertainty and the conclusion a reader might draw; then interpret the warranted interval and bound its population scope.

Video + follow-along: `u7_lesson6_live.html` (scan the code)
Finish (a)–(c) after the video. Turn in your sheet.